

Architecting Speech-Centric Large Language Models for Oral-Predominant Low-Resource Languages: A Strategic and Technical Blueprint for Dagbani

The proliferation of advanced natural language processing technologies and large language models has fundamentally revolutionized human-computer interaction, creating a new digital ecosystem driven by generative artificial intelligence. However, this transformation has predominantly favored high-resource, text-heavy languages, leaving a profound digital divide for linguistic communities whose cultures are rooted primarily in oral traditions.¹ For communities where oral communication supersedes written conventions, the deployment of a traditional text-in/text-out language model is fundamentally insufficient and practically exclusionary. Without integrated speech recognition and synthesis, oral archives, folktales, and everyday digital communications remain inaccessible, reinforcing digital hierarchies that prioritize written fluency over oral fluency.¹

To effectively serve these populations, the architectural paradigm must shift from traditional text-based systems to comprehensive speech-to-speech translation or conversational audio frameworks.³ The development of such technologies serves not only as a vital utility for digital inclusion but also as a mechanism for the preservation and revitalization of linguistic heritage.⁵ This comprehensive analysis provides a highly detailed, technical, and strategic blueprint for developing a fully functional, speech-enabled artificial intelligence ecosystem tailored to Dagbani. Dagbani is a deeply oral, tonal Gur language spoken predominantly in the Northern Region of Ghana.⁶ By integrating Automatic Speech Recognition (ASR), parameter-efficient large language model adaptation, and state-of-the-art Text-to-Speech (TTS) synthesis, researchers and software engineers can deploy localized, culturally resonant artificial intelligence systems that interface directly with native speakers through natural spoken dialogue.⁷

Sociolinguistic and Phonological Profiling of the Dagbani Language

The foundational and most critical step in developing any speech-enabled computational model is comprehensively mapping the target language's acoustic, morphological, and sociolinguistic topology. Artificial intelligence models designed for spoken interaction require rigorous acoustic parameters to decode and synthesize audio accurately. Dagbani, alternatively referred to as Dagbanli, belongs to the Maba subgroup of the Gur language family, which itself is a branch of the extensive Niger-Congo language phylum.⁹ The language is spoken by

approximately 1.16 million people, primarily comprising the Dagomba ethnic group.⁶ The linguistic and cultural epicenter of the Dagbamba people spans approximately twenty thousand square kilometers across northern Ghana, with the city of Yendi serving as the traditional and spiritual capital, and the metropolitan city of Tamale serving as the primary commercial and administrative hub.¹⁰

Developing automatic speech recognition and text-to-speech systems requires precise phoneme-to-grapheme and grapheme-to-phoneme mapping algorithms. Dagbani possesses a complex and highly specific phonemic inventory that computational models must be trained to recognize and reproduce. The language features eleven phonemic vowels, which are strictly categorized into six short vowels and five long vowels.¹² Furthermore, Dagbani exhibits five distinct nasal vowels, adding a layer of acoustic complexity to vowel articulation.¹³ The consonant inventory consists of eighteen primary consonants, including specialized labial-velar stops and a variety of fricatives and sonorants.¹²

Acoustic models must also account for well-attested allophonic variations that occur naturally in fluent speech. For instance, variation based on tongue-root advancement is prominent, leading to alternating pronunciations such as the shifting between close front unrounded vowels and their near-close equivalents.¹² Additionally, certain consonants undergo debuccalization depending on their phonetic environment; for example, specific alveolar fricatives may debuccalize into glottal fricatives when positioned between vowels, and certain voiced velar stops may reduce to glottal stops in post-vocalic positions.¹² When proceeding front vowels, several alveolar and velar sounds are realized as post-alveolar affricates or fricatives.¹²

Phonetic Category	Specific Dagbani Phonemes and Features
Short Vowels	a, e, i, o, u, ə (often transcribed as ɪ) ¹²
Long Vowels	a:, e:, i:, o:, u: ¹²
Nasal Vowels	an, ɛn, in, ɔn, un ¹³
Consonants	b, d, f, g, gb, h, j, k, l, m, n, p, r, s, ʃ, t, w, y ¹³
Allophonic Shifts	[i] ~ [ɪ], [e] ~ [ɛ], [u] ~ [ʊ], [o] ~ [ɔ] ¹²
Debuccalization	/s/ becomes [h] intervocalically; /g/ becomes [ʔ] post-vocalically ¹²

Dialectal variation represents another critical variable that severely impacts the robustness of

acoustic modeling. Dagbani features distinct regional dialects that are mutually intelligible but exhibit notable phonological divergences. The two most prominent dialects are Tòmòsílí, the Western dialect predominantly spoken in and around Tamale, and Nàyàhílí, the Eastern dialect spoken in Yendi and surrounding villages.⁹ Research into the phonological variation between these dialects reveals significant differences in vowel harmony systems; the Western dialect frequently utilizes advanced tongue root harmony, whereas the Eastern dialects predominantly employ retracted tongue root harmony.¹⁴ Furthermore, specific consonant alternations distinguish the dialects from one another.¹⁴ When engineering a speech recognition system, models trained exclusively on one dialect experience severe performance degradation when exposed to others due to these deeply ingrained acoustic differences.⁷ Consequently, data collection strategies must ensure balanced sampling across these geographic boundaries to prevent dialectal bias.

Beyond segmental phonology, the most computationally challenging aspect of Dagbani for artificial intelligence is its suprasegmental structure. Dagbani is a complex tonal language, utilizing three primary lexical tones to distinguish meaning: high, mid, and low.¹³ In the orthography, these tones are typically represented using an acute accent for the high tone, a grave accent for the low tone, and an optional macron for the mid tone.¹³ However, in everyday writing and digital communication, tone marks are frequently omitted entirely, forcing readers to rely heavily on syntactic and semantic context to disambiguate words.¹⁶ In tonal languages, even minute variations in pitch contour can completely alter the lexical meaning of a word or its grammatical function, leading to catastrophic misinterpretations by naive speech recognition systems.¹⁷

The tonal grammar of Dagbani extends far beyond simple lexical distinctions. The language exhibits complex morphological tone assignment rules where specific combinations of tense, aspect, and mood assign distinct tonal melodies to verb forms, often neutralizing the underlying tonal oppositions of the verb roots themselves.¹⁸ Furthermore, Dagbani exhibits postlexical phonology where tones alternate dynamically within and across noun phrases during continuous speech.¹⁸ In deliberate or slow speech patterns, speakers often insert weak epenthetic vowels between the coda of a first syllable and the onset of a second, producing completely open syllables on the surface.¹⁹ For a text-to-speech synthesis system to generate natural-sounding and comprehensible Dagbani speech, the underlying acoustic model must accurately predict these complex prosodic structures, metrical rules, and tonal contours from plain text that completely lacks diacritical markers.¹⁷

Data Landscape and Corpus Engineering Strategies

Machine learning algorithms are fundamentally bounded by the quality, diversity, and volume of their training data. For low-resource, predominantly oral languages like Dagbani, traditional data scraping techniques deployed for high-resource languages are entirely ineffective. Acquiring sufficient transcribed audio data requires novel crowdsourcing frameworks, highly controlled elicitation protocols, and extensive community mobilization.²¹ Fortunately, recent philanthropic and academic initiatives have significantly expanded the availability of specialized

Dagbani datasets, providing the necessary raw materials to train sophisticated speech models. One of the most monumental advancements in African language speech technology is the WAXAL project. Developed through extensive collaborations between Google, the University of Ghana, Makerere University, and other regional institutions, WAXAL provides a large-scale, openly accessible speech dataset spanning multiple African languages representing over one hundred million speakers.² The Ghanaian component of this initiative includes a massive five-thousand-hour speech corpus covering languages such as Akan, Ewe, Dagbani, Dagaare, and Ikposo.²³

The WAXAL dataset was meticulously engineered to overcome the common pitfalls of speech data collection in oral cultures. The dataset is bifurcated into two highly specialized streams to serve different phases of the machine learning pipeline. The first component, WAXAL-ASR, comprises spontaneous, unscripted speech.²³ Rather than asking participants to read standardized, highly formal text from a screen—a practice that often induces unnatural, stilted intonations—the researchers employed an image-prompted elicitation technique. Participants were shown visual stimuli covering dozens of everyday topics and asked to describe them naturally in their native language.²³ This methodology successfully captured authentic linguistic variations, natural tonal nuances, hesitation markers, and realistic code-switching, resulting in a dataset perfectly tuned for training robust speech recognition systems capable of understanding everyday conversation.²³ The second component, WAXAL-TTS, contains over two hundred hours of studio-grade, single-speaker recordings across the project's languages.² These recordings involve professional speakers reading phonetically balanced scripts in controlled acoustic environments, providing the pristine, noise-free audio required for training high-fidelity voice synthesis models.²

Beyond academic-led data collection, community-driven crowdsourcing has proven to be a highly effective and scalable methodology in low-resource environments.⁵ The Dagbani Wikimedians User Group, a collective of language activists and educators based primarily in Tamale, has actively driven the collection of voice data through the Mozilla Common Voice initiative.⁷ Following extensive training on digital activism and online safety, these volunteers mobilized their communities to donate voice recordings.²⁶ As of the version 24 release in early 2026, the Dagbani dataset on the Common Voice platform contains over forty thousand validated written sentences and more than twenty thousand corresponding audio recordings.²⁵ Nearly a quarter of these recordings include detailed metadata regarding the speaker's gender, age, and specific regional dialect, providing invaluable demographic features for machine learning engineers seeking to build unbiased acoustic models.²⁵ The community volunteers also manage the rigorous annotation and validation workflows, scrutinizing the data for linguistic correctness, cultural relevance, and pronunciation consistency before it is approved for open-source release.²⁷

Researchers have also successfully compiled customized, highly targeted datasets using localized mobile applications designed specifically to overcome hardware limitations. A notable data collection pipeline utilized the Spell4Wiki Android application to record thousands of short word lists and sentences sourced directly from the Dagbani Wikipedia.⁷ Because professional

recording studios are often inaccessible in rural northern Ghana, recordings were captured in residential homes using standard smartphones and consumer-grade earphones, representing the exact acoustic environments where the final application will likely be used.⁷ The methodology deliberately mixed sentences from both the Eastern and Western dialects to ensure the resulting model would be robust against geographic pronunciation shifts.⁷ To ensure quality control, a team of five native-speaker proofreaders verified that every audio recording precisely matched its textual transcription pair.⁷ This grassroots effort yielded a highly specific corpus of over eleven thousand utterances, equating to approximately nine and a half hours of aligned audio, which was subsequently hosted on Wikimedia Commons under a permissive public domain license.⁷

To train the underlying large language model for translation, reasoning, and semantic understanding, massive parallel text corpora are absolutely essential. The GhanaNLP initiative has systematically addressed this gap by curating an extensive, high-quality dataset of over forty-one thousand parallel sentence pairs aligning local Ghanaian languages—including Dagbani—with English.²⁸ These datasets were not scraped automatically but were meticulously collected, translated, and annotated by human linguistic professionals.²⁸ The data is enriched with standard structural metadata to ensure consistency and is explicitly designed to support downstream commercial and educational applications, providing the foundational semantic mapping necessary for accurate text-to-text generative tasks.²⁸

Dataset Designation	Primary Modality	Approximate Volume	Unique Characteristics and Collection Methodology	Source
WAXAL-ASR	Audio + Text Transcript	Subset of 5,000 Ghanaian hours	Image-prompted, spontaneous, unscripted speech capturing natural code-switching.	²
WAXAL-TTS	High-Fidelity Audio	235 total hours across project	Studio-quality, single-speaker, phonetically balanced scripts for	²²

			synthesis.	
Mozilla Common Voice	Audio + Text Transcript	20,000+ validated recordings	Crowdsourced read-speech, extensive demographic metadata, community-validated.	25
Spell4Wiki Corpus	Audio + Text Transcript	11,207 utterances (~9.5 hours)	Smartphone-recorded in natural environments, deliberate mixing of regional dialects.	7
GhanaNLP Parallel	Text	41,513 parallel sentence pairs	Professional human translations between English and local Ghanaian languages.	28

Architecting the Automatic Speech Recognition (ASR) Pipeline

The first computational node in a fully integrated, speech-enabled language model system is the Automatic Speech Recognition module. This system acts as the auditory cortex of the application, responsible for decoding the user's spoken Dagbani into a highly accurate textual representation that the core language model can subsequently process and understand. Architecting this system for an underrepresented, tonal language involves navigating extreme data scarcity and complex morphophonological rules.

Historically, speech recognition systems relied on heavily engineered pipelines utilizing Hidden Markov Models combined with Gaussian Mixture Models.³⁰ These classical architectures required exhaustive, manually crafted phonetic dictionaries, explicit acoustic feature extraction, and complex decision trees to function.³⁰ However, the artificial intelligence industry has

experienced a paradigm shift toward deep neural networks and self-supervised learning frameworks, fundamentally altering how low-resource language processing is approached.¹ Self-supervised acoustic models, such as Facebook's Wav2Vec 2.0, OpenAI's Whisper, and the Massively Multilingual Speech framework, leverage vast quantities of unlabeled, raw audio data from across the globe to learn universal, generalized representations of human speech mechanics.⁴ By pre-training on thousands of hours of diverse linguistic sounds, these models develop a foundational understanding of acoustics. Once this foundational understanding is established, the models can be fine-tuned using a remarkably small dataset of transcribed Dagbani audio to achieve astonishingly high accuracy.⁷

In documented empirical experiments, researchers successfully engineered a functional ASR system for Dagbani by fine-tuning pre-trained Wav2Vec 2.0 checkpoints utilizing the Hugging Face Transformers library.⁷ The researchers selected the wav2vec2-large-xlsr-53 checkpoint, a massive model that had been extensively pre-trained on fifty-three distinct global languages.⁷ They partitioned their collected Dagbani dataset into a ninety-five percent training split and a five percent validation split.⁷ Through rigorous optimization, the model reached convergence in merely nine training epochs, taking approximately two hours of compute time on a standard consumer-grade graphical processing unit.⁷ The resulting fine-tuned system achieved a highly impressive Word Error Rate of 0.347.⁷ This experiment conclusively proved that massive self-supervised models, when properly calibrated, can rapidly adapt to the unique acoustic signatures of Dagbani with less than ten hours of labeled training data.⁷

When native transcribed data is severely constrained, cross-lingual transfer learning emerges as an indispensable optimization strategy.³³ This approach involves pre-training a neural acoustic model on a high-resource language, or a collection of geographically proximate language families, before subsequently adapting the neural weights to the target language.³⁴ Research demonstrates that transferring the hidden layers from neural networks trained on phonologically related languages significantly reduces the ultimate Word Error Rate of the target system.²¹

The specific architecture chosen for cross-lingual transfer plays a critical role in preserving accuracy. Empirical evaluations contrasting global multilingual models, such as multilingual BERT, against regionally focused models reveal stark differences in performance.³³ Regionally specialized architectures, such as the Africa-centric AfriBERT or the language-family-specific BantuBERTa, exhibit significantly superior cross-lingual transfer capabilities when navigating African linguistic structures.³³ In benchmark tests for cross-lingual transfer, AfriBERT achieved an impressive accuracy rate of 88.3 percent post-fine-tuning, drastically outperforming generic global models.³³ Furthermore, massive large-scale pre-training is proven to be essential for algorithmic robustness; models like AfriBERT exhibit minimal catastrophic forgetting—a phenomenon where a neural network abruptly forgets previously learned information upon learning new data—whereas traditional architectures suffer severe performance degradation during the fine-tuning phase.³³ For the Dagbani context, initializing an acoustic model that has been heavily pre-trained on closely related West African Gur languages provides a highly stabilized starting point, reducing the necessary fine-tuning data

requirements by several orders of magnitude.³

A persistent and severe challenge in Dagbani speech recognition stems from the lack of universal orthographic standardization across the geographic region.⁷ While an official orthography was approved by the Dagbani Orthography Committee in the late twentieth century, its adoption in everyday digital writing is highly inconsistent.⁷ Textual datasets frequently contain multiple, equally valid spelling variations for identical words.⁷ Consequently, a standard grapheme-level speech recognition model might accurately decode the spoken phonetic sequence but output a spelling variant that differs from the specific ground truth in the training dataset, artificially and incorrectly inflating the reported Word Error Rate.⁷

To comprehensively mitigate this statistical anomaly, adopting a phoneme-level acoustic architecture rather than a grapheme-level system is highly recommended for unwritten or orthographically fluid languages.³⁷ Phoneme-level evaluation utilizes the International Phonetic Alphabet to directly and unambiguously map acoustic features to distinct linguistic sounds, bypassing spelling conventions entirely.³⁸ Advanced open-source tools like Epitran can systematically generate precise phonemic transcriptions from messy textual data, bridging the gap between acoustic realities and non-standardized text.³⁸ Phoneme-level speech recognition ensures the model is graded on acoustic accuracy rather than spelling adherence, and more importantly, it allows the architecture to more accurately capture and represent the vital tonal contours inherent in Dagbani speech that are entirely invisible in standard orthography.³⁰

Parameter-Efficient Large Language Model (LLM) Adaptation and Textual Reasoning

Once the automatic speech recognition module successfully decodes the spoken input into a textual or phonemic representation, the core large language model takes over to execute the required semantic reasoning, translation, summarization, or conversational logic. Training a monolithic foundation model entirely from scratch exclusively for a single low-resource language is computationally prohibitive, financially unfeasible, and ultimately data-inefficient.³⁹ Instead, the standard industry protocol involves adapting existing, highly capable open-weight models through advanced Parameter-Efficient Fine-Tuning techniques.

State-of-the-art pre-trained models, such as Meta's Llama 3.1 architecture, exhibit extraordinary generalized reasoning capabilities, vast world knowledge, and strong instruction-following behaviors. However, they require highly targeted, rigorous alignment to accurately process Dagbani syntax, comprehend cultural idioms, and respect local knowledge structures.⁴¹ A highly effective and economically viable technique for this alignment is Low-Rank Adaptation. This technique fundamentally alters the fine-tuning process by freezing the vast majority of the pre-trained model's original dense neural weights and injecting small, trainable rank decomposition matrices directly into the transformer architecture.⁴² This sophisticated mathematical approach reduces the total number of actively trainable parameters by a factor of ten thousand, allowing developers to execute complex fine-tuning operations on standard, consumer-grade hardware rather than relying on expensive

supercomputing clusters.

An existing, highly successful implementation available on the Hugging Face repository demonstrates the practical application of this technique by adapting the unsloth/Meta-Llama-3.1-8B-Instruct model specifically for bidirectional English-to-Dagbani translation tasks.⁴² The developer utilized a highly optimized 4-bit Low-Rank Adaptation configuration to maximize efficiency while preserving semantic depth. The specific hyperparameter configuration utilized an adaptation rank of sixteen and an alpha scaling factor of sixteen.⁴² Crucially, the technique targeted all major attention projection matrices within the transformer layers, specifically the query, key, value, and output projections, alongside the corresponding gate, up, and down projections in the feed-forward network.⁴² The optimization process employed an 8-bit AdamW optimizer with a learning rate set precisely at 0.0002.⁴² This meticulous configuration enables the language model to strictly follow user instructions and generate highly accurate, culturally context-aware Dagbani outputs without suffering catastrophic degradation of its foundational English reasoning capabilities or its overarching logical frameworks.⁴²

However, given the unique infrastructural constraints prevalent in rural Northern Ghana—including intermittent power grids, expensive cellular data costs, and severe bandwidth limitations—deploying massive, cloud-dependent language models may prove highly impractical due to crippling latency.⁴³ Therefore, optimizing the language model for localized, on-device inference at the edge is a vital strategic imperative.

Apple's Open Efficient Language Model framework provides a robust architectural blueprint specifically designed for extreme resource-constrained environments.⁴⁴ Unlike standard transformer architectures that distribute parameters completely uniformly across all internal layers, this framework employs an innovative layer-wise scaling strategy. This technique allocates parameters asymmetrically, strategically concentrating computational power in the layers where it yields the highest semantic accuracy, thereby maximizing performance within a strictly restricted parameter budget. Empirical testing demonstrates that a heavily constrained one-billion-parameter model using this layer-wise scaling strategy achieved a 2.36 percent accuracy improvement over standard uniform models, while simultaneously requiring fifty percent fewer pre-training tokens to reach convergence.⁴⁴ These highly efficient models can be seamlessly converted via specialized libraries for rapid, low-latency execution directly on mobile edge devices.⁴⁴

Furthermore, Liquid Foundation Models, pioneered by MIT spinoff Liquid AI, offer an even more radical departure from conventional architectures.³⁹ These models utilize dynamic, non-transformer liquid neural networks that require substantially fewer neurons and drastically reduce the operational memory footprint, making them the ultimate solution for decentralized edge computing.³⁹ Models within this ecosystem are engineered to operate smoothly and rapidly on standard central processing units or mobile neural processing units.³⁹ Deploying a fine-tuned liquid neural network as the central cognitive engine for a Dagbani application would guarantee real-time conversational responsiveness, completely independent of cloud connectivity or cellular data availability.³⁹

Text-to-Speech (TTS) Synthesis and Generative Voice Architecture

The final operational node in the speech-enabled ecosystem converts the language model's textual or phonetic output back into a natural, fluid, and highly comprehensible spoken Dagbani acoustic signal. Text-to-speech synthesis is arguably the most mathematically and computationally complex component to engineer for low-resource tonal languages. The underlying acoustic model must autonomously infer incredibly complex prosody, pitch fluctuations, and temporal rhythm from plain text that completely lacks explicit markers for these vital acoustic features.¹⁷

Modern voice generation systems utilize sophisticated encoder-decoder neural architectures, representing a complete departure from traditional Statistical Parametric Speech Synthesis frameworks, which infamously produced highly robotic, emotionally flat, and artificially over-smoothed audio waveforms.²⁰ Contemporary neural vocoders operate by mapping abstract phonemic representations directly to high-fidelity, highly dynamic audio waveforms.²⁰ The Massively Multilingual Speech text-to-speech framework, developed by researchers at Facebook, utilizes the highly acclaimed VITS architecture, which stands for Variational Inference with adversarial learning for end-to-end Text-to-Speech.⁴⁶ This architecture functions as an immensely powerful conditional variational autoencoder comprising three primary components: a posterior encoder, a decoder, and a conditional prior.⁴⁶ A highly advanced text encoder and multiple flow-based coupling layers work in tandem to predict complex spectrogram features, which are subsequently decoded into pristine audio waveforms utilizing an adversarial HiFi-GAN vocoder.⁴⁶ Crucially for languages like Dagbani, the VITS architecture intrinsically incorporates a stochastic duration predictor alongside normalizing flows.⁴⁶ This allows the model to mathematically account for the one-to-many nature of human speech, meaning the system can synthesize highly dynamic audio featuring incredibly diverse rhythms, natural pauses, and shifting intonations from the exact same textual input sequence.⁴⁶ While pre-trained VITS checkpoints exist for other African languages, a bespoke architecture can be rapidly trained from scratch for Dagbani by utilizing the pristine, studio-quality recordings found in the WAXAL-TTS dataset.²²

The absolute bleeding edge of audio synthesis technology entirely bypasses traditional, modular text-to-speech pipelines in favor of unified, large language model-based universal audio foundation models. The UniAudio architecture represents a massive paradigm shift away from narrow, task-specific synthesis engines.⁴⁷ Scaled to over one billion parameters and rigorously trained on an unprecedented 165,000 hours of highly diverse acoustic data, UniAudio leverages the exact same next-token prediction mathematical techniques utilized by text-based language models to generate high-fidelity speech.⁴⁷

The operational mechanism of UniAudio is elegantly simple yet incredibly powerful. First, the model tokenizes the target audio utilizing a complex residual vector quantization-based neural codec, effectively turning sound waves into a discrete, finite vocabulary of audio tokens.⁴⁷

Second, it concatenates the input conditions—such as textual descriptions or phonemic sequences—directly with the audio tokens into one continuous, unified sequence.⁴⁷ Finally, a massive multi-scale Transformer architecture analyzes the sequence and predicts the subsequent continuous audio tokens, generating the final sound.⁴⁷ Because the UniAudio framework mathematically treats audio generation as a standard language modeling task, it possesses immense cross-lingual generalization capabilities and can seamlessly support completely novel languages via simple parameter fine-tuning.⁴⁹

A highly advanced and practically essential feature required for deploying a user-facing conversational artificial intelligence agent is the ability to maintain a deeply consistent persona and highly natural voice quality over extended interactions. Models such as Chatterbox Turbo offer robust, open-source solutions for high-fidelity generative speech synthesis.⁵² Chatterbox Turbo fundamentally leverages a universal phonemic space, systematically mapping phonetic similarities across completely divergent language families.⁵³ It utilizes a deeply pre-trained speaker encoder network capable of extracting complex vocal characteristics from incredibly brief reference audio clips, enabling pristine zero-shot voice cloning capabilities.⁵³

By extensively employing cross-lingual transfer learning algorithms, the Chatterbox architecture effectively borrows deep speech intuition—encompassing complex prosody, breathing mechanics, and emotional inflection—from high-resource languages like English or Mandarin, and applies this acoustic knowledge to low-resource languages that possess minimal training data.⁵³ Crucially from a deployment perspective, the highly permissive MIT-licensed open-source nature of the Chatterbox framework allows software engineers to self-host the entire model locally on independent graphical processing units. This bypasses the crippling latency delays, strict usage limitations, and prohibitive financial costs associated with closed-source commercial application programming interfaces operated by entities like ElevenLabs or OpenAI.⁵²

Integrating Paradigms: End-to-End versus Cascaded Speech-to-Speech Translation

To seamlessly synthesize the speech recognition, language model, and voice generation components into a fluid, user-facing application, system architects must definitively choose between deploying a cascaded framework or a unified end-to-end model.³

A classic cascaded speech-to-speech translation system sequentially links the individual models in a rigid, linear pipeline. The user's spoken Dagbani is routed through the ASR module, converted to text, processed by the core language model for translation or reasoning, outputted as Dagbani text, and finally routed through the TTS module to generate the artificial speech response.

The primary, undeniable advantage of a cascaded architecture is its extreme modularity.⁴ Each distinct component within the pipeline can be aggressively optimized, meticulously debugged, and seamlessly updated completely independently of the others.⁴ For instance, engineers can effortlessly swap a standard, heavy Transformer language model for a highly efficient OpenELM architecture without fundamentally disrupting the functionality of the ASR or TTS

modules.⁴⁴ Furthermore, cascaded systems benefit instantaneously from any independent advancements in text-based language modeling or specialized adversarial vocoders.⁵⁴ Rigorous academic evaluations, such as the comprehensive NaijaS2ST benchmark—which exhaustively tested translation models across complex linguistic contexts including Nigerian Pidgin, Hausa, Igbo, and Yoruba—conclusively demonstrated that combining specialized AudioLLMs with targeted few-shot examples within a cascaded framework remains highly robust, frequently outperforming highly complex end-to-end systems in difficult translation tasks.⁵⁵

The primary, often crippling drawback of the cascading methodology is severe latency accumulation. The distinct computational processing time required for the ASR, the LLM, and the TTS models aggressively compounds sequentially, frequently resulting in massive delays that completely shatter the illusion of a fluid, real-time human-computer conversation. Conversely, end-to-end models mathematically bypass intermediate textual representations entirely, directly mapping the source audio spectrogram of the user to the target audio spectrogram of the response.³ Early iterations of end-to-end models, such as the Translatotron architecture, pioneered this highly integrated approach, demonstrating massive reductions in inference latency.³

However, constructing highly accurate end-to-end systems demands astronomically vast amounts of perfectly aligned parallel audio data, a resource that is practically nonexistent for low-resource languages like Dagbani.³ To creatively circumvent this extreme data scarcity, researchers heavily utilize auxiliary structural transfer strategies. By rigorously pre-training a massive, multi-task end-to-end model on highly abundant language pairs—such as French to English—and subsequently transferring the deeply learned encoder parameters to the low-resource language pair, engineers can achieve highly significant improvements in translation accuracy and structural coherence.³ Comprehensive ablation studies reveal that mathematically transferring the acoustic encoder provides exponentially greater performance gains than attempting to transfer the decoder modules or relying on auxiliary speech recognition modules.³

As the artificial intelligence industry relentlessly matures, native multimodal architectures that process and generate raw acoustic tokens directly—such as Google's sophisticated Gemini 3.0 framework—will inevitably become the undisputed industry standard. These unified architectures will gracefully bridge the widening gap between the crippling latency of cascaded systems and the impossible data demands of traditional end-to-end models.⁵⁷

Deployment Infrastructure and Ecosystem Integration

Constructing highly accurate, culturally resonant artificial intelligence models is only half of the overarching engineering challenge. Deploying these sophisticated systems effectively to end-users in rural or developing regions requires incredibly robust backend orchestration and highly optimized delivery infrastructure.

Hosting and reliably serving fine-tuned large language models requires specialized, high-performance infrastructure designed to maintain strict uptime service level agreements

and ensure highly predictable, low-latency inference. Enterprise-grade platforms, such as the Nebius Token Factory, provide highly secure, dedicated cloud environments specifically engineered for deploying custom-tuned checkpoints of major open-source models.⁵⁸ These sophisticated platforms offer a suite of advanced features including aggressive dynamic autoscaling to handle unpredictable traffic spikes, multi-region geographic routing to ensure consistent performance regardless of user location, and highly complex speculative decoding algorithms designed specifically to drastically accelerate inference speeds.⁵⁸ Utilizing such robust, enterprise-grade deployment infrastructure allows software developers and language activists to focus their limited resources entirely on critical data curation and precise model tuning, rather than becoming bogged down in relentless backend server maintenance and security patching.⁵⁸

To integrate the custom Dagbani artificial intelligence seamlessly into highly practical, everyday applications—such as automated WhatsApp advisory bots, interactive educational software, or civic communication platforms—the underlying machine learning models must be elegantly wrapped in universally accessible Application Programming Interfaces (APIs). The GhanaNLP initiative provides a highly functional, industry-standard template through its comprehensive Khaya AI Translation API platform.²⁸

The sophisticated Khaya platform exposes entirely distinct, highly optimized REST endpoints for specific tasks, including Text Translation, Automatic Speech Recognition, and Text-to-Speech synthesis.⁵⁹ By utilizing a universally standard JSON request payload containing the specific input strings and the highly exact ISO language codes (for example, utilizing the parameter `lang: "en-dag"` to denote an English to Dagbani translation request), external software developers can seamlessly and rapidly inject deep Dagbani linguistic intelligence into completely third-party digital interfaces.⁵⁹ By architecting the custom Dagbani model securely behind a standardized REST or high-speed WebSocket API framework, the complex technology can be rapidly democratized and aggressively adopted by local, grassroots software engineers.²⁸

A paramount and unavoidable consideration for any technological deployment in Northern Ghana is the severe unreliability of internet connectivity and the highly prohibitive cost of cellular data.⁴³ Architecting an application that requires constant, high-bandwidth data streams for raw audio transmission to remote cloud servers will automatically exclude a massive, highly vulnerable portion of the target demographic.⁴³

Strategies to effectively mitigate this profound infrastructural barrier include aggressive optimization for on-device processing. By ruthlessly utilizing highly quantized, parameter-efficient models—specifically leveraging the layer-wise scaling of OpenELM or the dynamic memory footprint of Liquid Foundation Models—software engineers can force the entire inference process to run directly on the user's local smartphone hardware.³⁹ This fundamentally eliminates the absolute necessity for constant cloud connectivity, allowing the speech system to function flawlessly in entirely offline environments.³⁹

An alternative, highly impactful strategy involves aggressive corporate partnerships to secure zero-rated network access. This involves collaborating directly with massive regional telecommunications providers and global tech conglomerates to completely zero-rate the

cellular data associated specifically with the artificial intelligence application. For example, recent high-level initiatives involving Google's heavy investment in Ghana have resulted in groundbreaking agreements to deploy vital educational AI tools on a strictly zero-rated basis.⁶² This guarantees that end-users, regardless of their socioeconomic status or geographic isolation, do not incur any prohibitive data charges when accessing crucial, curriculum-aligned artificial intelligence content, ensuring absolute digital inclusivity.⁶²

Strategic Blueprint for Dagbani Speech AI

The engineering and deployment of a fully integrated, highly accurate large language model and conversational speech interface for a predominantly oral language like Dagbani is an incredibly intricate, highly multidisciplinary, but ultimately achievable objective. The necessary transition from an exclusionary, text-centric artificial intelligence paradigm to a highly inclusive, audio-native ecosystem requires the meticulous orchestration of diverse technologies, ranging from acoustic phonetics and sociolinguistics to advanced neural vocoding and edge-computing infrastructure.

To successfully execute this monumental project, researchers, software engineers, and language activists must adhere to a highly rigorous, multi-phased strategic blueprint. First, engineers must aggressively leverage existing, validated open-source audio corpora rather than attempting to build massive datasets entirely from scratch. Projects must extensively utilize the spontaneous, image-prompted recordings of the WAXAL dataset, the highly validated, community-driven archives of Mozilla Common Voice, and the dialectically diverse Spell4Wiki corpus to acquire the tens of thousands of specific Dagbani utterances strictly required for foundational acoustic training.²

Second, the development of the Automatic Speech Recognition pipeline must heavily implement advanced cross-lingual transfer learning protocols. Engineers should utilize highly proven self-supervised architectures such as Wav2Vec 2.0 or the Massively Multilingual Speech framework. Crucially, the training sequence must be initialized utilizing checkpoints that have been heavily pre-trained on phonologically similar Gur languages or broader, regionally focused African language families—such as the AfriBERT architecture—to effectively bypass the crippling cold-start data scarcity problem and completely avoid catastrophic forgetting.³

Third, developers must strictly adopt rigorous phoneme-level modeling methodologies. To effectively counteract the severe lack of regional orthographic standardization, and to accurately capture and mathematically represent Dagbani's vital high, mid, and low tonal registers, the ASR system must map raw acoustic features directly to the International Phonetic Alphabet using tools like Epitran, rather than relying on highly ambiguous text characters.³⁷

Fourth, the core cognitive reasoning engine must be developed by fine-tuning highly efficient large language models via Low-Rank Adaptation. Engineers should utilize Parameter-Efficient Fine-Tuning techniques on powerful open-weight models, trained exhaustively on highly accurate parallel English-Dagbani text corpora, to ensure the resulting system deeply understands cultural context, follows strict instructions, and respects local syntax.²⁸

Fifth, the system must utilize next-generation Text-to-Speech neural vocoders to generate realistic audio. Developers should employ advanced architectures like VITS or highly scaled

universal audio foundation models such as UniAudio and Chatterbox Turbo. These incredibly sophisticated systems mathematically utilize stochastic duration predictors and universal, cross-lingual phoneme representations to generate highly natural, emotionally resonant, and rhythmically accurate spoken output.⁴⁶

Finally, the deployment architecture must prioritize a cascaded, highly edge-optimized system framework. Initially deploying a cascaded speech-to-speech translation system guarantees absolute modularity and allows for rapid, independent component upgrades. By specifically utilizing layer-wise scaled models or dynamic liquid neural networks, engineers can enable incredibly low-latency, entirely on-device inference, ensuring the artificial intelligence remains fully accessible in deeply rural environments plagued by highly unstable internet connectivity.⁴ By strictly adhering to this comprehensive, highly technical blueprint, developers can utterly transcend the fundamental limitations of text-only artificial intelligence. This approach guarantees the delivery of a highly sophisticated, culturally aware conversational AI ecosystem that actively preserves the deep oral traditions of the Dagbamba people while aggressively democratizing access to the absolute bleeding edge of computational reasoning.

Works cited

1. Omnilingual ASR: Open-Source Multilingual Speech Recognition for 1600+ Languages, accessed May 1, 2026, <https://arxiv.org/html/2511.09690v1>
2. WAXAL: A Large-Scale Multilingual African Language Speech Corpus - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2602.02734v2>
3. Improving Speech-to-Speech Translation for Low-Resource Languages via Transfer Learning - APSIPA, accessed May 1, 2026, http://www.apsipa.org/proceedings/2025/papers/APSIPA2025_P208.pdf
4. Contributing to Speech-to-Speech Translation for African Low-Resource Languages : Study of French-Mooré Pair - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/2026.loreslm-1.54.pdf>
5. Cultivating Spoken Language Technologies for Unwritten Languages | Request PDF, accessed May 1, 2026, https://www.researchgate.net/publication/380522310_Cultivating_Spoken_Language_Technologies_for_Unwritten_Languages
6. Q&A: Meet Issahaku Tidoo Abass, Dagbani-language activist and Wikimedian from Ghana, accessed May 1, 2026, <https://rising.globalvoices.org/blog/2025/04/11/qa-meet-issahaku-tidoo-abass-dagbani-language-activist-and-wikimedian-from-ghana/>
7. BREAKING THE LOW-RESOURCE BARRIER FOR DAGBANI ASR: FROM DATA COLLECTION TO ASR MODELING - OpenReview, accessed May 1, 2026, <https://openreview.net/pdf?id=lje9II9zV8>
8. NoLoR: An ASR-Based Framework for Expedited Endangered Language Documentation with Neo-Aramaic as a Case Study - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2412.04717v1>
9. Dagbani English The Influence of Dagbani on the Use of English in Ghana Memunatu Sheini - Epub Bayreuth, accessed May 1, 2026,

- <https://epub.uni-bayreuth.de/6513/1/Thesis%20Submitted%20Revised%2029%20June%20PDF%202.pdf>
10. Culture & History Archives - Page 2 of 3 - Ghana Goods, accessed May 1, 2026, <https://ghanagoods.co.uk/category/ghana/ghana-culture-history/page/2/>
 11. Grandmasters of the Drum: A Literary Linguistic Analysis of the Dagbamba Panegyrics, accessed May 1, 2026, https://www.academia.edu/95676973/Grandmasters_of_the_Drum_A_Literary_Linguistic_Analysis_of_the_Dagbamba_Panegyrics
 12. Dagbani language - Wikipedia, accessed May 1, 2026, https://en.wikipedia.org/wiki/Dagbani_language
 13. Bridging Gaps in Natural Language Processing for Yorùbá: A Systematic Review of a Decade of Progress and Prospects - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2502.17364v1>
 14. (PDF) Phonological Variation in Dagbani Dialects - ResearchGate, accessed May 1, 2026, https://www.researchgate.net/publication/355007425_Phonological_Variation_in_Dagbani_Dialects
 15. Building ASR Resources for the Hutsul Dialect of Ukrainian - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/2026.vardial-1.15.pdf>
 16. Afrocentric NLP for African Languages | PDF | Tone (Linguistics) | Orthography - Scribd, accessed May 1, 2026, <https://www.scribd.com/document/946275849/Towards-Afrocentric-NLP-for-African-Languages-Wher>
 17. Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2505.11690v1>
 18. Dagbani Verb Tonology - Berkeley Linguistics, accessed May 1, 2026, https://www.linguistics.berkeley.edu/~hyman/Dagbani_Boston_5.1_final.pdf
 19. Defining the prosodic word with segmental processes in Dagbani in - AKJournals, accessed May 1, 2026, <https://akjournals.com/view/journals/2062/69/3/article-p271.xml>
 20. Advancing Text-to-Speech Systems for Low-Resource Languages: Challenges, Innovations, and Future Directions - IEEE Xplore, accessed May 1, 2026, <https://ieeexplore.ieee.org/iel8/6287639/10820123/11146779.pdf>
 21. Building Large-Vocabulary ASR Systems for Languages Without Any Audio Training Data - ISCA Archive, accessed May 1, 2026, https://www.isca-archive.org/interspeech_2019/prasad19_interspeech.pdf
 22. [2602.02734] WAXAL: A Large-Scale Multilingual African Language Speech Corpus - arXiv, accessed May 1, 2026, <https://arxiv.org/abs/2602.02734>
 23. WAXAL: A large-scale open resource for African language speech technology, accessed May 1, 2026, <https://research.google/blog/waxal-a-large-scale-open-resource-for-african-language-speech-technology/>
 24. Introducing WAXAL: A New Open Dataset for African Speech Technology - Google Blog, accessed May 1, 2026,

- <https://blog.google/intl/en-africa/company-news/outreach-and-initiatives/introducing-waxal-a-new-open-dataset-for-african-speech-technology/>
25. Mozilla Common Voice Meets Wikidata (How the Dagbanli Dictionary Got Audio Usage Examples) - Wikimedia Diff, accessed May 1, 2026, <https://diff.wikimedia.org/2026/04/22/mozilla-common-voice-meets-wikidata-how-the-dagbanli-dictionary-got-audio-usage-examples/>
 26. Dagbani Language Activists Trained to Promote and Protect Their Work Online, accessed May 1, 2026, <https://diff.wikimedia.org/2025/06/05/dagbani-language-activists-trained-to-promote-and-protect-their-work-online/>
 27. Dagbani Community Builds Voice AI Dataset in Tamale | Let's Data Science, accessed May 1, 2026, <https://letsdatascience.com/news/dagbani-community-builds-voice-ai-dataset-in-tamale-e7fcd19>
 28. GhanaNLP Parallel Corpora: Comprehensive Multilingual Resources for Low-Resource Ghanaian Languages - ResearchGate, accessed May 1, 2026, https://www.researchgate.net/publication/402480817_GhanaNLP_Parallel_Corpora_Comprehensive_Multilingual_Resources_for_Low-Resource_Ghanaian_Languages
 29. Latest Common Voice Dataset Surpasses 20,000 Hours of Open-Source Speech Data - Mozilla Foundation, accessed May 1, 2026, <https://www.mozillafoundation.org/en/blog/latest-common-voice-dataset-surpasses-20000-hours-of-open-source-speech-data/>
 30. Improving phoneme set discovery for documenting unwritten languages - Machine Learning Lab, accessed May 1, 2026, https://ml.informatik.uni-freiburg.de/wp-content/uploads/papers/17_ESSV.pdf
 31. TOWARDS PHONEME INVENTORY DISCOVERY FOR DOCUMENTATION OF UNWRITTEN LANGUAGES Markus Müller, Jörg Franke, Alex Waibel, Sebastia - KIT, accessed May 1, 2026, <https://isl.iar.kit.edu/pdf/Mueller2017a.pdf>
 32. Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/2025.africanlp-1.13.pdf>
 33. Cross-lingual transfer of multilingual models on low resource African Languages - sacair 2025, accessed May 1, 2026, https://2025.sacair.org.za/online-proceedings/Papers/paper_87.pdf
 34. Cross-lingual transfer of multilingual models on low resource African Languages - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2409.10965v1>
 35. Transfer Learning for Less-Resourced Semitic Languages Speech Recognition: the Case of Amharic - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/2020.sltu-1.9.pdf>
 36. Automatic Speech Recognition (ASR) for African Low-Resource Languages: A Systematic Literature Review - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2510.01145v1>
 37. (PDF) Hard to Be Heard: Phoneme-Level ASR Analysis of Phonologically Complex, Low-Resource Endangered Languages - ResearchGate, accessed May 1, 2026,

https://www.researchgate.net/publication/404021382_Hard_to_Be_Heard_Phone-me-Level_ASR_Analysis_of_Phonologically_Complex_Low-Resource_Endangered_Languages

38. Breaking Language Barriers Through ASR Implementation with TensorFlow | by Sciforce, accessed May 1, 2026, <https://medium.com/sciforce/breaking-language-barriers-through-asr-implementation-with-tensorflow-0000f7a8e10a>
39. Liquid AI: Build efficient general-purpose AI at every scale., accessed May 1, 2026, <https://www.liquid.ai/>
40. OpenAI's 'o3' Has Programmers Fearful, <https://mail.google.com/mail/u/0/#all/FMfcgzQXKqkcCSvXxhbNWXcNgSQqmvSM>
41. Workshop on African Natural Language Processing (AfricaNLP) (2026) - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/events/africanlp-2026/>
42. abdulhafis/dagbani-english-translator - Hugging Face, accessed May 1, 2026, <https://huggingface.co/abdulhafis/dagbani-english-translator>
43. The Push to Digitize Ghana's Indigenous Languages - Susinsight, accessed May 1, 2026, <https://susinsight.com/ghana-indigenous-language-tech>
44. OpenELM: An Efficient Language Model Family with Open Training ..., accessed May 1, 2026, <https://machinelearning.apple.com/research/openelm>
45. Next OpenAI Model To Do "5hr Tasks", <https://mail.google.com/mail/u/0/#all/FMfcgzQXJQVmfDTcBmpBGnHWKQcZSsfq>
46. facebook/mms-tts-tir - Hugging Face, accessed May 1, 2026, <https://huggingface.co/facebook/mms-tts-tir>
47. UNIAUDIO: AN AUDIO FOUNDATION MODEL TOWARD UNIVERSAL AUDIO GENERATION - OpenReview, accessed May 1, 2026, <https://openreview.net/pdf?id=nhgTmx1TZJ>
48. UniAudio: Towards Universal Audio Generation with Large Language Models, accessed May 1, 2026, <https://www1.se.cuhk.edu.hk/~hccl/publications/pub/ICML2024.pdf>
49. UniAudio: Towards Universal Audio Generation with Large ..., accessed May 1, 2026, https://uniaudio666.github.io/demo_UniAudio/
50. [2310.00704] UniAudio: An Audio Foundation Model Toward Universal Audio Generation - arXiv, accessed May 1, 2026, <https://arxiv.org/abs/2310.00704>
51. UniAudio: An Audio Foundation Model Toward Universal Audio Generation - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2310.00704v5>
52. Chatterbox: Open Source Text-to-Speech | Resemble AI, accessed May 1, 2026, <https://www.resemble.ai/chatterbox/>
53. accessed May 1, 2026, https://www.youtube.com/watch?v=p-NL_s2X-VY
54. Speech-to-Speech Translation Pipelines for Conversations in Low-Resource Languages - ACL Anthology, accessed May 1, 2026, <https://aclanthology.org/2025.mtsummit-2.3.pdf>
55. NaijaS2ST: A Multi-Accent Benchmark for Speech-to-Speech Translation in Low-Resource Nigerian Languages - arXiv, accessed May 1, 2026, <https://arxiv.org/html/2604.16287v1>
56. Projects - MATRIX Lab - BYU, accessed May 1, 2026,

- <https://matrix.byu.edu/Projects>
57. Google's Gemini 3.0 Confirmed,
<https://mail.google.com/mail/u/0/#all/FMfcgzQcqQxGZggpZtdccPntTTcNKNd>
 58. Nebius Token Factory, accessed May 1, 2026,
<https://nebius.com/services/token-factory>
 59. Home - Khaya AI API - developer portal, accessed May 1, 2026,
<https://translation.ghananlp.org/>
 60. APIs: List - Khaya AI API - developer portal - Ghana NLP, accessed May 1, 2026,
<https://translation.ghananlp.org/apis>
 61. Khaya African Language Translation and Speech Recognition AI Demonstrates Major Improvements | by Paul Azunre, accessed May 1, 2026,
<https://azunre.medium.com/khaya-african-language-translation-and-speech-recognition-ai-demonstrates-major-improvements-e13d9fb26fb8>
 62. Ghana to roll out AI education tools in local languages, accessed May 1, 2026,
<https://www.modernghana.com/news/1465780/ghana-to-roll-out-ai-education-tools-in-local-lang.html>
 63. Ghana introduces AI tools in Twi, Ewe, Dagbani to expand education - CitiNewsroom.com, accessed May 1, 2026,
<https://citinewsroom.com/2026/01/ghana-introduces-ai-tools-in-twi-ewe-dagbani-to-expand-education/>